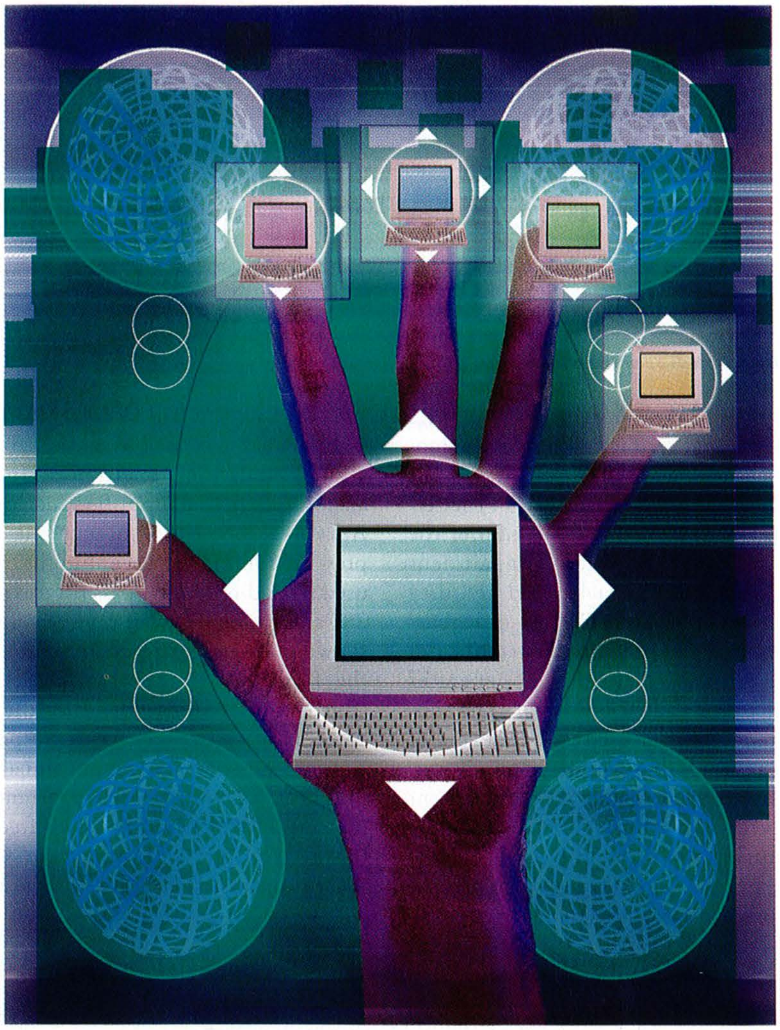

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Current Measurements in Digitally Controlled AC Drives

Seung-Ho Song, Jong-Woo Choi, and Seung-Ki Sul

In modern digital drives, all of the control functions are implemented in software. One such function is the current regulator, which is an essential part of an ac motor drive system. Because the performance of field-oriented control greatly depends on the performance of the regulator, there have been many studies to improve the performance of the regulator [1]-[5]. But relatively few works are reported to obtain accurate feedback current through an interface circuit consisting of signal conditioners and an A/D converter [6]-[8].

Pulse-width-modulation (PWM) operation of an inverter inherently produces harmonic voltages along with the desired fundamental voltages. These, in turn, cause harmonic currents [8], [9], which can be picked up by the measuring equipment. The direct utilization of

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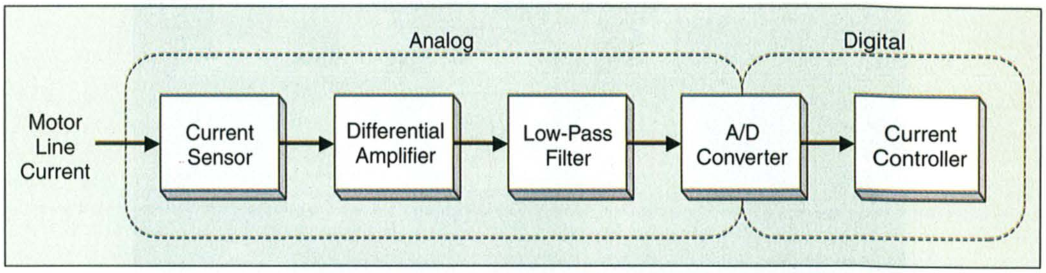

Fig. 1. Sampling procedure of currents.

the measurement values containing harmonics degrades the control characteristics in the motor control system. The consequences of high-order harmonics are distortion in the reference voltages, lower permissible gain of the current regulators with resultant impairment of dynamic behavior, and a limited stability range of the control systems. The low-order harmonics contained in measured currents degrades field-oriented control and generates torque ripple in the induction motor. The control system requires measured values devoid of the inherent inverter-induced harmonics.

Efforts to avoid the harmonics in measurement can be grouped into two categories. One is the method of instantaneous sampling [6] and the other is obtaining locally averaged values [7]. In the case of constant switching frequency space vector PWM (SVPWM), [6] suggests that it is possible to detect the fundamental component of a current by sampling the current at the midpoint of the zero space vectors in a PWM pulse pattern. Reference [8] proposed the sampling of current twice symmetrically before and after the midpoint to achieve simultaneous sampling of I_a and I_b. That paper also compares the current control bandwidth of averaged sampling with that of instantaneous sampling. However, instantaneously sampled values contain some errors with a low-pass filter, which is inevitably used for the suppression of switching noise. Averaged sampling requires additional hardware for implementation and results in one step delay.

In this article, we investigate the effect of the low-pass filter on current sampling and we propose a technique to compensate the delay of the filter. The current-sampling error due to the time delay is calculated analytically from the voltage reference vector in three-phase symmetry SVPWM. The effects of this current-sampling error, such as improper torque regulation and degradation of control performance, are investigated and the effectiveness of the proposed technique are verified through simulations and experiments.

Characteristic Of Low-Pass Filter

The current feedback is obtained through the signal-conditioning circuits, including the analog differential amplifier and low-pass filter. A typical procedure of current sampling is presented in Fig. 1. A configuration of a widely used second-order low-pass filter is shown in Fig. 2(a) and its frequency response using Bode plot is shown in (b) and (c), where the cutoff frequency of the filter is 5.11 kHz with critical damping (Butterworth type). Note that the phase angle delay is converted to the corresponding delay time as a function of the input signal frequency in Fig. 2(c), showing a nearly constant value up to the cutoff frequency. The cutoff frequency of the low-pass filter is designed to suppress the switching noise. The higher the cutoff frequency,

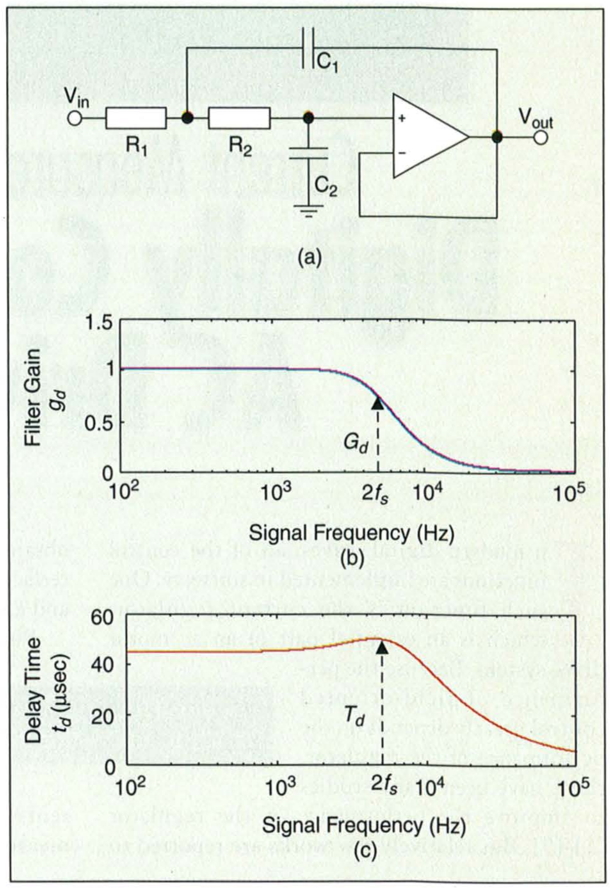

Fig. 2. Characteristics of a second-order low-pass filter: (a) configuration, (b) gain, and (c) delay time due to phase lag.

the lower the constant value of delay time is extended to the longer range of frequency. The effect of anti-aliasing usually cannot be obtained in high-performance ac drives because a cutoff frequency lower than half the PWM frequency would cause a significant time delay, comparable to the control period. In the case of symmetrical SVPWM, the dominant harmonic frequency of actual current is twice the switching frequency, $2f_s$. To simplify the analysis of the filtering effect, the low-pass filter is modeled to have constant gain, G_d, and delay time, T_d, of the frequency of $2f_s$ as follows:

$$V_{out}(t) \approx G_d V_{in}(t - T_d). \quad (1)$$

PWM Strategy and Current Sampling

There are eight available states in the space vector representation of the three machine voltages. Two zero vectors and two effective voltage vectors are used to synthesize the reference voltage vector in

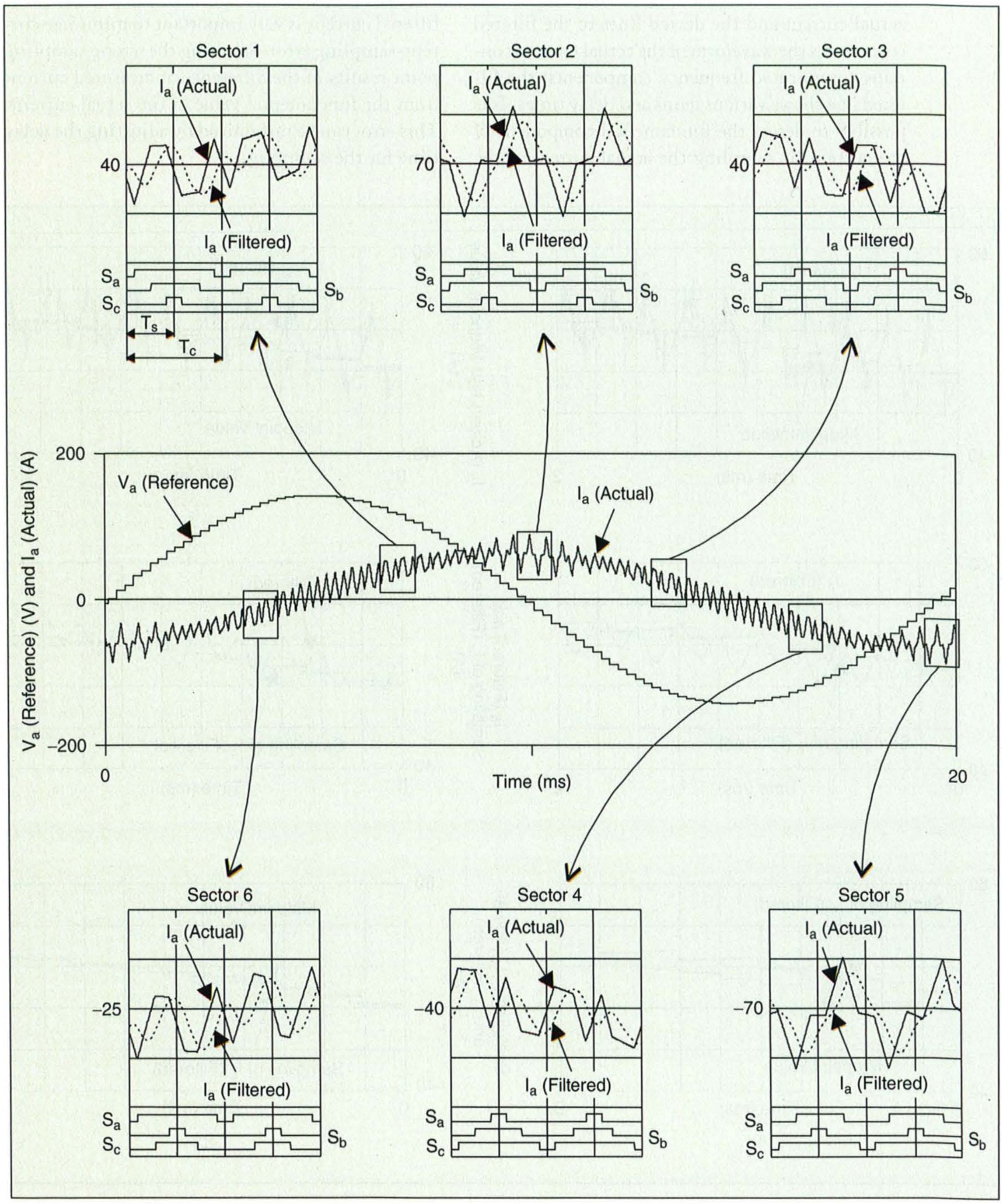

Fig. 3. PWM signals, motor current, and filtered current at each sector during 50-Hz operation (the switching frequency is 2.5 kHz and the filter cutoff frequency is 5.1 kHz).

SVPWM strategy. To obtain the optimum pulse pattern, the switching sequence must be arranged symmetrically in such a way that the transition from one state to the next is performed by switching only one inverter leg [9]. Fig. 3 shows the ripple current in each sector according to the three-phase space vector PWM switching pattern [10]. The current sampling and control algorithms are performed at the period of T_s (frequency of $2f_s$) to reduce current ripple and to achieve fast response of tracking in many high-performance digital ac drives. The solid lines correspond to the actual current and the dotted lines to the filtered current. As the waveform of the actual current contains complicated frequency components, the filtered one shows various gains and delay times. It is possible to detect the fundamental component of the current by sampling the actual current at the midpoint of the zero space vectors in a PWM pulse pattern. When the sampling and control routine is performed twice in a switching period T_c, the sampling of actual current at the midpoint of each zero vector can be regarded as the fundamental component. However, the motor current is delayed as the current is sampled through a low-pass filter, and the sampled current at the midpoint of the zero vector is not the fundamental component. The current-sampling error is defined to be the difference between the sampled current and the fundamental one. The proper choice of the sampling point of the filtered current is very important to minimize current-sampling error. Selecting the wrong sampling point results in the difference of measured current from the fundamental value of the actual current. This error can be minimized by adjusting the delay time for the sampling.

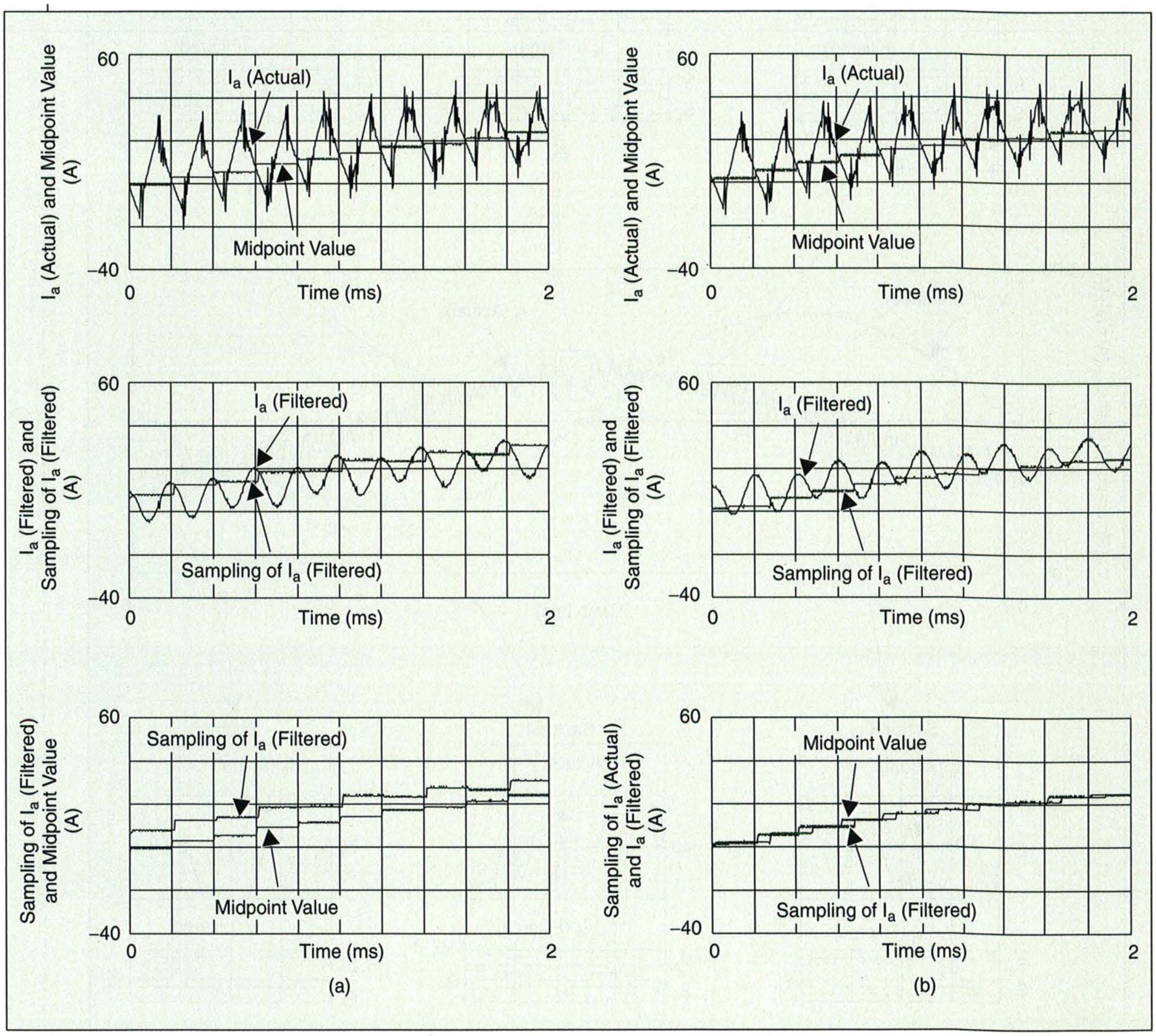

Fig. 4. Sampling of actual motor current and filtered current (experiment): (a) without delayed sampling, (b) with 90° delayed sampling.

In a real experimental setup, Fig. 4 shows the actual motor current, sampling point, filtered current, and sampled current at typical operating conditions. In Fig. 4(a), the actual current and the midpoint value are drawn in the top. The actual current waveform is obtained by an isolated current measurement system and contains a lot of noise mainly due to the patristics with the switching of power devices. The second graph shows the filtered current and its sampling at the same midpoint of the zero vector without considering filter delay. The difference between the sampling of filtered current and the midpoint value of actual current, as compared in the last plot of Fig. 4(a), shows an oscillatory characteristic as the ripple current shapes in "on" and "off" modes of PWM period are quite different. Fig. 4(b) shows the case of delayed

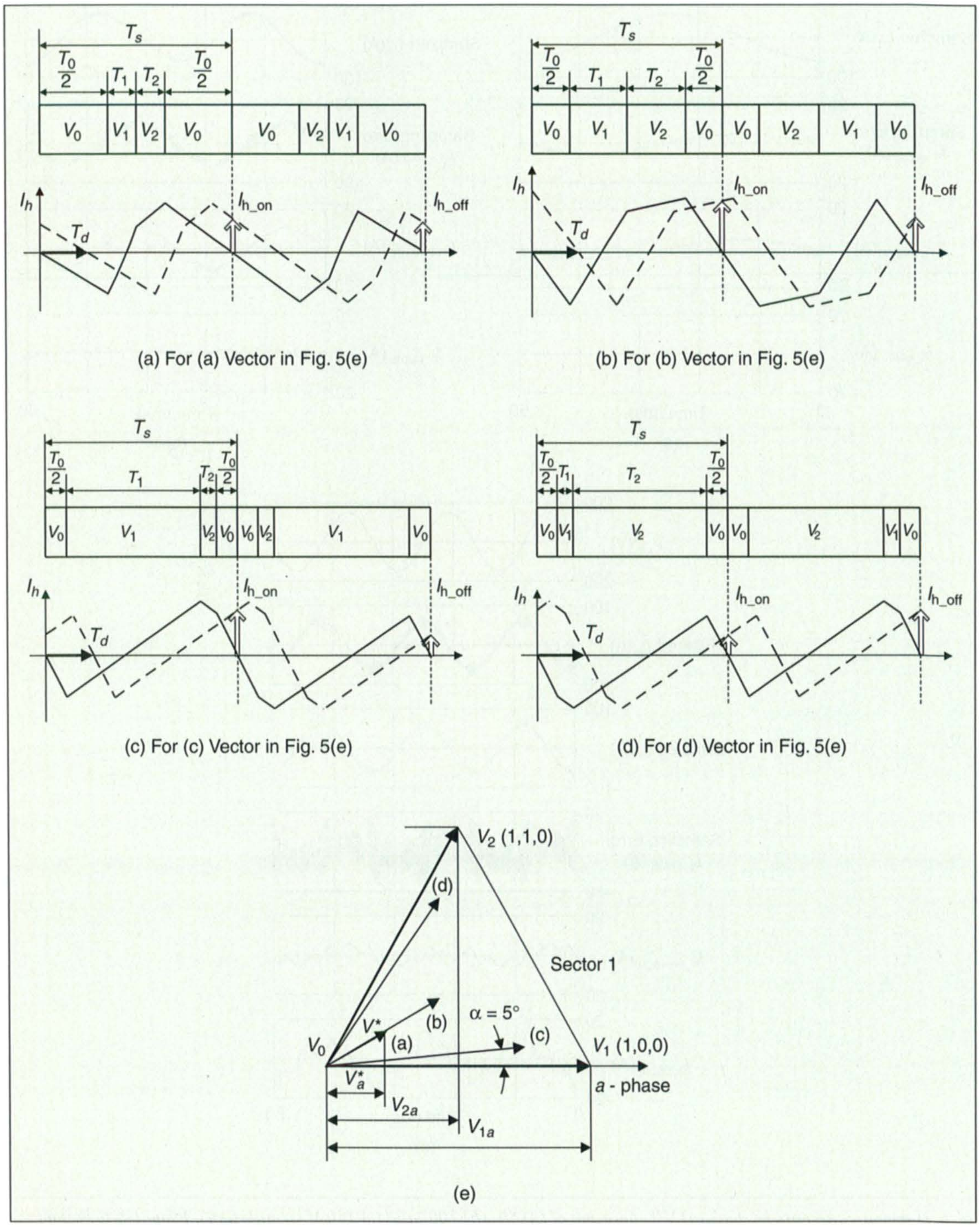

Fig. 5. Equivalent sampling of harmonic currents in four cases of voltage vector: (a) $|V^*| = 50$ V, $\alpha = 30°$, *(b)* $|V^*| = 100$ V, $\alpha = 30°$, *(c)* $|V^*| = 150$ V, $\alpha = 5°$, *(d)* $|V^*| = 150$ V, $\alpha = 55°$, *and (e) reference voltage vectors.*

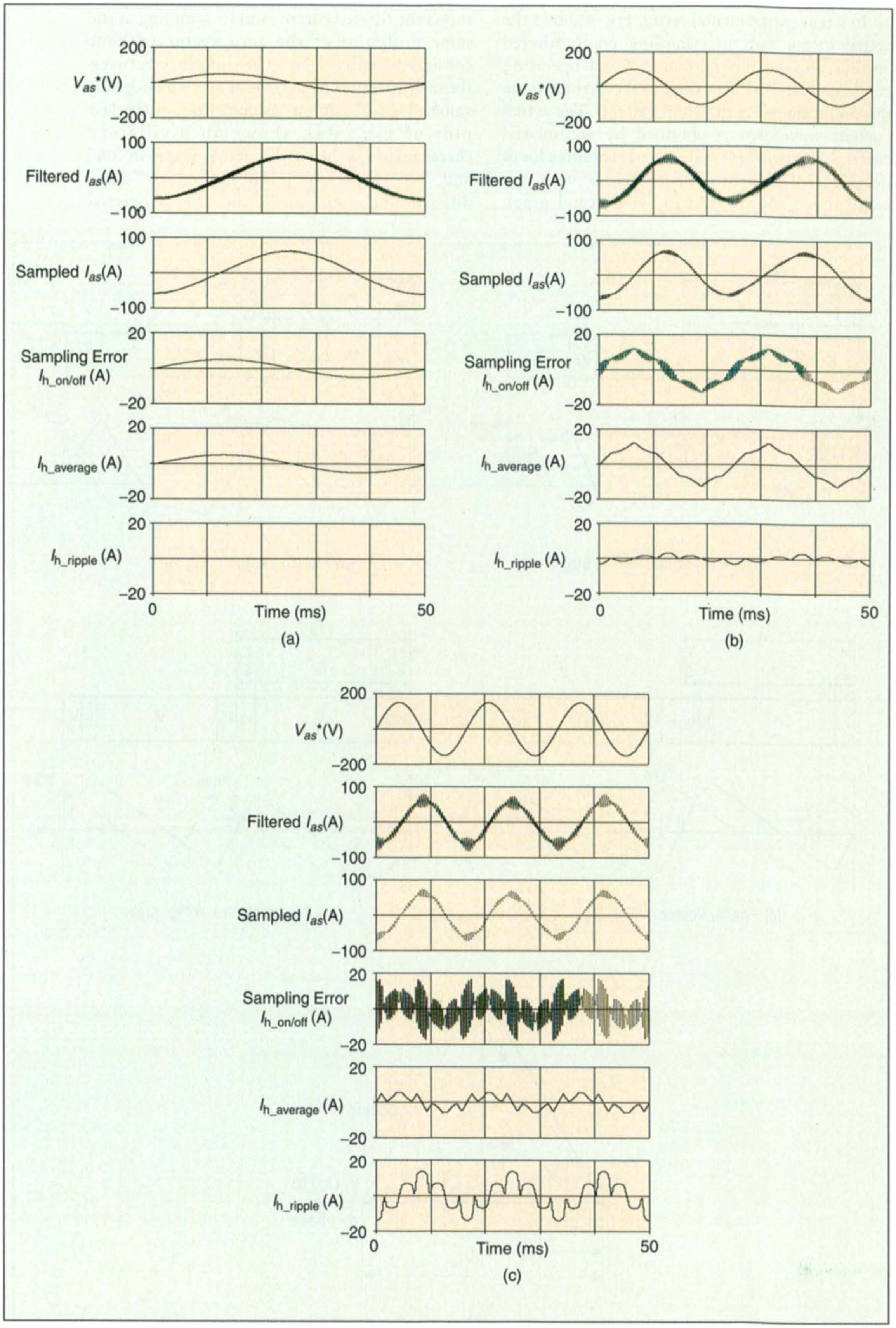

Fig. 6. Harmonic currents in constant V/f operation of (a) 50, (b) 100, and (c) 150 V (simulation). From top to bottom, a-phase voltage references, filtered a-phase currents, sampled a-phase currents, calculated current-sampling error, average of the error, and ripple of the error.

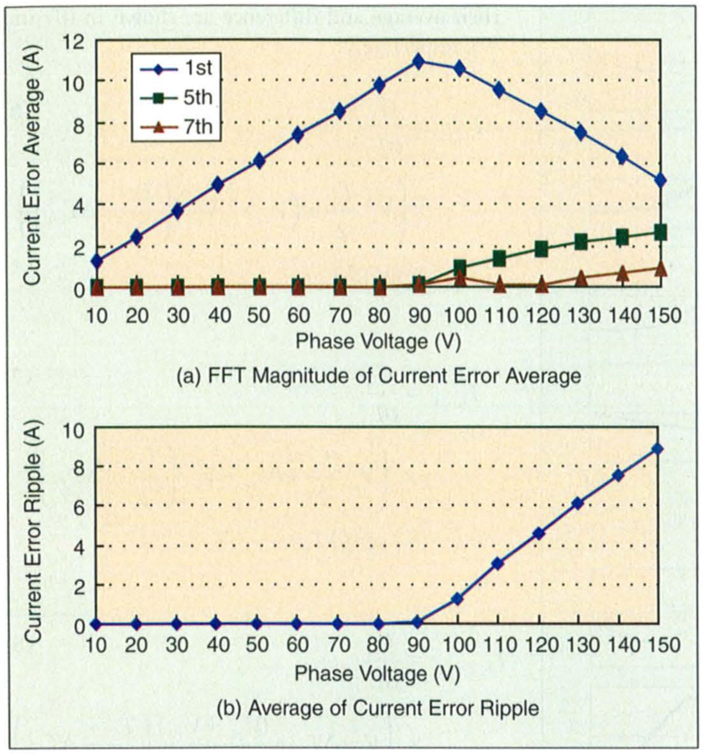

Fig. 7. Current-sampling error against phase voltage without sampling delay. (a) FFT result of current error average, $I_{h\_average}$. (b) Average of current error ripple, I_{h_ripple}.

sation delay, T_{ds}, is assumed to be zero in Fig. 5 in which the sampling of filtered current without compensation corresponds to the sampling of original current with the amount of $\Delta T_d = T_d$ in advance.

A. Case 1: "On" and "Off" Sample during Zero Vector

In the case of longer zero vector time than delay time error i.e., $|\Delta T_d| < \frac{T_0}{2}$, the sampled harmonic currents at "on" and "off" modes have the same value because the voltage applied during zero vector is the same. So their average, $I_{h\_average}$, has the same value as $I_{h\_on} = I_{h\_off}$ and depends on the filter gain and delay time error as shown in (3). Graphical representations of these harmonic currents are shown in Fig. 5(a) with corresponding PWM patterns.

sampling of filtered current. Here, sampled current shows good agreement with the midpoint value with time delay. The amount of delay time is a constant 50 •, which is nearly 90° at the filter cutoff frequency.

Analysis of Current-Sampling Error

If the filtered current is sampled with delay time, T_{ds} from the midpoint of zero vector, the time error between real delay, T_d, and compensation delay, T_{ds}, can be expressed as follows:

$$\Delta T_d = T_d - T_{ds}. \qquad (2)$$

The current-sampling error due to this timing error can be calculated directly from the voltage reference and PWM variables based on the following assumptions:

- The voltage drop at stator resistance is so negligible that all the voltage is applied on the leakage inductance σL_s.
- The phase voltage reference is constant during the sampling period as $V_a^* = (V_{1a}T_1 + V_{2a}T_2)/T_s$.
- The filter gain G_d and delay time T_d are constant as shown in (1).

In Fig. 5(e), four typical voltage reference vectors, which represent four cases of Fig. 5(a), (b), (c), and (d), respectively, are given. The compen-

$$I_{h\_on} = I_{h\_off} = G_d \times \frac{V_a^* \times |\Delta T_d|}{\sigma L_s} \times \mathrm{sgn}(\Delta T_d) \qquad (3)$$

where

I_{h_on}, I_{h_off} = a-phase current-sampling error calculated at each "on" and "off" mode;

G_d = filter gain at the frequency of $2f_s$;

σL_s = the equivalent leakage inductance;

V_a^* = a-phase voltage reference.

$$I_{h\_average} = G_d \times \frac{V_a^* \times |\Delta T_d|}{\sigma L_s} \times \mathrm{sgn}(\Delta T_d). \qquad (4)$$

$$I_{h\_ripple} = 0. \qquad (5)$$

B. Case 2: "On" (or "Off") Sample during First Effective Vector

On the other hand, in the case of Fig. 5(b), where $|\Delta T_d| < \frac{T_0}{2}$, the sampled harmonic current of the first "on" mode, I_{h_on}, is different from that of the second "off" mode, I_{h_off}, due to the difference in the effective voltage applied. The harmonic currents can be calculated as shown in (6) and (7), and

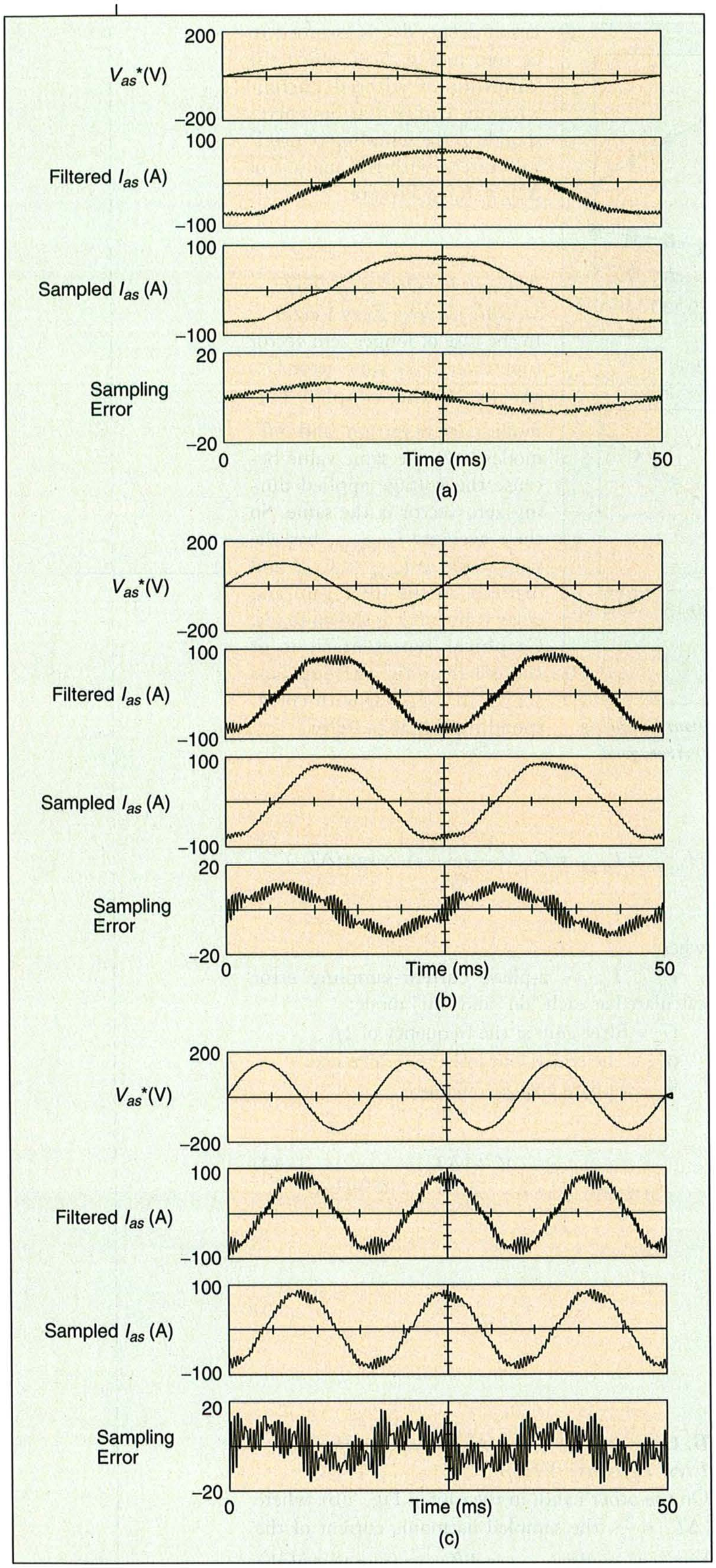

Fig. 8. Harmonic currents in constant V/f operation of (a) 50, (b) 100, and (c) 150 V (experiment). From top to bottom, a-phase voltage references, filtered a-phase currents, sampled a-phase currents, and measured current-sampling error.

their average and difference are shown in (8) and (9), respectively.

$$I_{\text{h\_on}} = \frac{G_d}{\sigma L_s} \times \left(V_a^* \frac{T_0}{2} + \left(V_{2a} - V_a^*\right)\left(\frac{T_0}{2} - |\Delta T_d|\right)\right) \times \operatorname{sgn}(\Delta T_d). \tag{6}$$

$$I_{\text{h\_off}} = \frac{G_d}{\sigma L_s} \times \left(V_a^* \frac{T_0}{2} + (V_{1a} - V_a^*)\left(\frac{T_0}{2} - |\Delta T_d|\right)\right) \times \operatorname{sgn}(\Delta T_d). \tag{7}$$

$$I_{\text{h\_average}} = \frac{G_d}{\sigma L_s} \times \left(V_a^* |\Delta T_d| + \frac{(V_{2a} + V_{1a})}{2}\left(\frac{T_0}{2} - |\Delta T_d|\right)\right) \times \operatorname{sgn}(\Delta T_d). \tag{8}$$

$$I_{\text{h\_ripple}} = \frac{G_d}{\sigma L_s} \times (V_{2a} - V_{1a})\left(\frac{T_0}{2} - |\Delta T_d|\right) \times \operatorname{sgn}(\Delta T_d). \tag{9}$$

C. Case 3: "On" Sample during Second Effective Vector

Fig. 5(c) where $|\Delta T_d| > \frac{T_0}{2} + T_2$ and Fig. 5(d) where $|\Delta T_d| > \frac{T_0}{2} + T_1$ usually occur when a large reference voltage vector is located near the basis vector as shown in Fig. 5(e). Equations (10) and (11) are the harmonic currents in the case of (c). Their averages and differences are shown in (12) and (13), respectively.

$$I_{\text{h\_on}} = \frac{G_d}{\sigma L_s} \times \left(V_a^* \frac{T_0}{2} + (V_{2a} - V_a^*)\left(\frac{T_0}{2} - |\Delta T_d|\right)\right). \times \operatorname{sgn}(\Delta T_d) \tag{10}$$

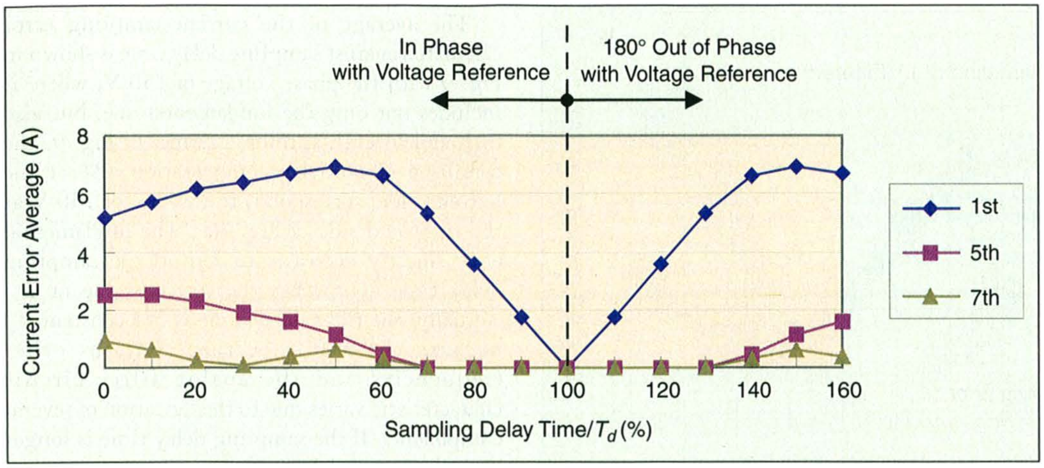

Fig. 9. Current-sampling error against sampling delay time.

$$I_{h\_off} = \frac{G_d}{\sigma L_s} \times \left(V_a^* \frac{T_0}{2} + (V_{1a} - V_a^*)(-T_1) + (V_{2a} - V_a^*) \left(\frac{T_0}{2} - |\Delta T_d| + T_1 \right) \right) \times \mathrm{sgn}(\Delta T_d). \tag{11}$$

$$I_{h\_average} = \frac{G_d}{\sigma L_s} \times (V_a^* |\Delta T_d| + V_{2a} (\tfrac{T_0}{2} - |\Delta T_d| + \tfrac{T_1}{2}) - V_{1a} \tfrac{T_1}{2}) \times \mathrm{sgn}(\Delta T_d). \tag{12}$$

$$I_{h\_ripple} = \frac{G_d}{\sigma L_s} \times (V_{1a} - V_{2a}) \times T_1 \times \mathrm{sgn}(\Delta T_d). \tag{13}$$

D. Case 4: "Off" Sample during Second Effective Vector

The case of Fig. 5(d) is similar to the case of (c), and the average and ripple current error are given by

$$I_{h\_average} = \frac{G_d}{\sigma L_s} \times (V_a^* |\Delta T_d| + V_{1a} (\tfrac{T_0}{2} - |\Delta T_d| + \tfrac{T_2}{2}) - V_{2a} \tfrac{T_2}{2} \times \mathrm{sgn}(\Delta T_d) \tag{14}$$

and

$$I_{h\_ripple} = \frac{G_d}{\sigma L_s} \times (V_{1a} - V_{2a}) \times T_2 \times \mathrm{sgn}(\Delta T_d). \tag{15}$$

The simulation results without delay compensation are given in Fig. 6 with the reference voltages of (a) 50 V, (b) 100 V, and (c) 150 V in a constant V/f pattern. A-phase voltage references, filtered a-phase currents, sampled a-phase currents, calculated harmonic currents, averages of harmonic currents, and differences of harmonic currents are shown from top to bottom. The average of the harmonic currents, $I_{h\_average}$ is in phase with the phase voltage reference and contains not only fundamental, but also higher-order harmonic terms in Fig. 6(c) as the voltage reference increases. I_{h_ripple}, the difference between harmonic currents I_{h_on} and I_{h_off}, corresponds to the ripple magnitude of harmonic currents. I_{h_ripple} remains zero until the zero vector time is longer than the delay time error, i.e., $|\Delta T_d| < \frac{T_0}{2}$. It has two peaks in every fundamental period in sectors 2 and 5 where the magnitude of $V_{1a} - V_{2a}$ is twice the others.

In Fig. 7(a), a result of the FFT analysis of $I_{h\_average}$ is plotted against the phase voltage. This low-order current-sampling error causes a disagreement between the generated and reference torque. The average of I_{h_ripple} changes linearly, as shown in Fig. 7(b), when the reference voltage exceeds a certain level. This increment of ripple in the sampled current reduces the maximum bandwidth of current control.

Experimental waveforms are given in Fig. 8 under the same conditions as the simulation. These results are very similar to that of the simulation. A current-sampling error plotted at the bottom is the difference of sampled currents with and without filtering. The distortion of current waveforms comes from nonlinearities such as dead time and zero current clamping effects in actual situations. The current-sampling error at 50 V contains small ripples, which seem to be the result of incomplete filter modeling. The current-sampling error due to the delay effect of the low-pass filter become more severe as the voltage increases.

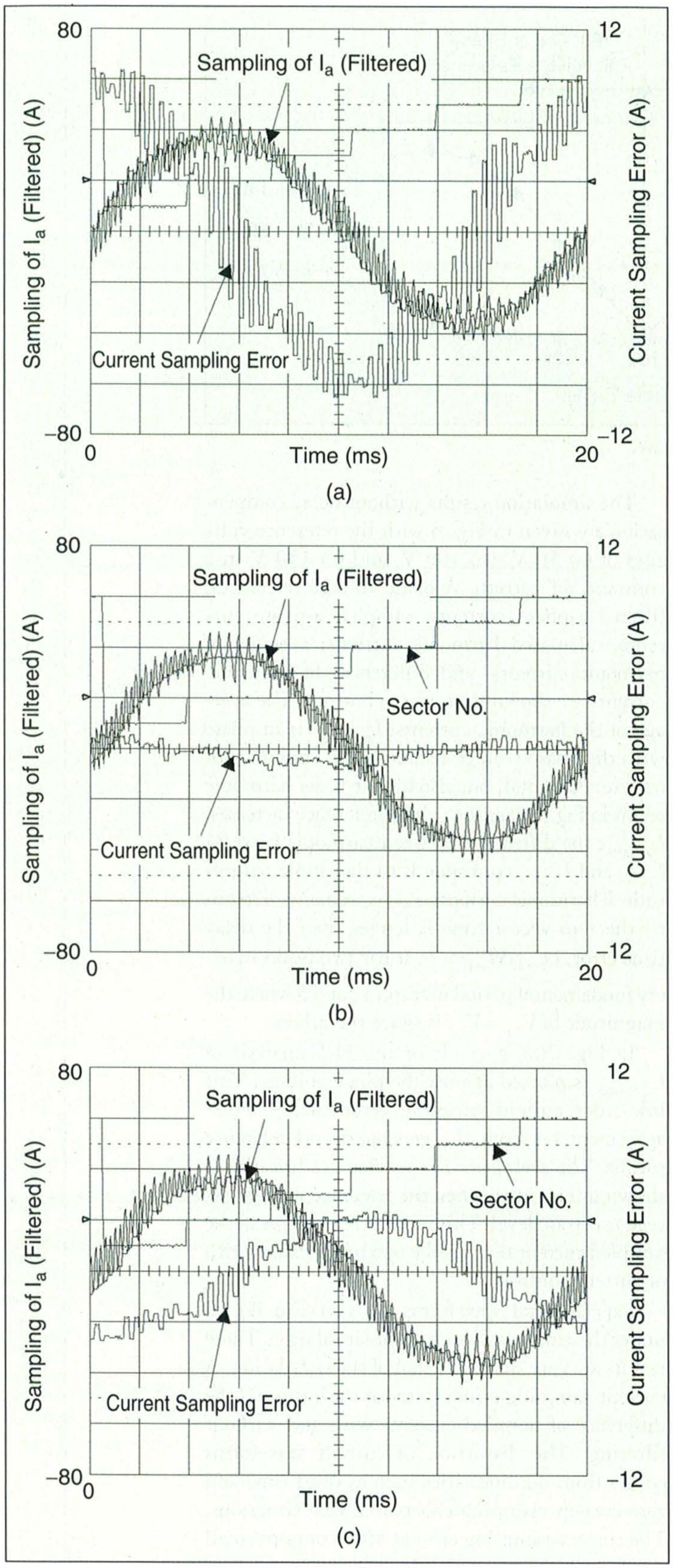

Fig. 10. Sampling of filtered current and the current-sampling error at 50-Hz operation (experiment) (a) without delayed sampling, (b) with 100% delayed sampling, and (c) with 140% delayed sampling.

The average of the current-sampling error calculated against sampling delay time is shown in Fig. 9 with the phase voltage of 150 V, where it includes not only the fundamental one, but also fifth and seventh harmonic terms. In Fig. 9, the condition of no-delay compensation, 0% at the left-end side, corresponds to the case of 150 V at the right-end side of Fig. 7(a). The fundamental error linearly decreases to zero as the sampling delay time approaches the correct value of T_d. Actually, the filter delay time is not constant T_d because the ripple current contains other frequencies and the analog filter circuit characteristic varies due to the deviation of several components. If the sampling delay time is longer than the actual delay, the sampling error has the same magnitude symmetrical to the 100% point and it becomes 180° out of phase with the phase-voltage reference. The experimental waveforms show the same results in Fig. 10. Fig. 10(a) is the case of no compensation without delayed sampling during 50-Hz operation. The analog filtered current and the sampled current are plotted in the same scale and the current-sampling error is composed of a fundamental and ripple term. The average of the current error is in phase with the a-phase voltage, and the ripple term oscillates at the switching frequency. The sector numbers from one to six are included for synchronous triggering with phase voltage reference. Fig. 10(b) is the case of 100% delayed sampling, and (c) is the case of 140% delayed sampling in which the current error is out of phase with phase voltage.

The phase and magnitude error of the measured current degrades the performance of the field orientation control of the machine drive system. The dynamic response is tested to show a torque regulation failure due to the current-sampling error, which is confirmed by the simulation in Fig. 11 and by the experiment in Fig. 12 when the speed reference was changed from 0 to 1500 rpm. In the simulation in Fig. 11, the motor speed, the reference/feedback of torque current, the current-sampling error, and the reference/real torque are plotted from top to bottom. The motor currents are sampled without delay in Fig. 11(a), and sampled with a delay time of 50 μs, which corresponds to the filter delay time T_d, in Fig. 11(b). The average value of the generated torque is less than the reference torque without delayed sampling. On the other hand, the average value of the generated torque is nearly the same as the reference torque with delayed sampling, which is shown in Fig. 11(b), resulting in a 9% reduction of acceleration time compared to the case of undelayed sampling. The results of the experiment in Fig. 12 coincide with that of simulation with reduction of acceleration time of 10.2 ms.

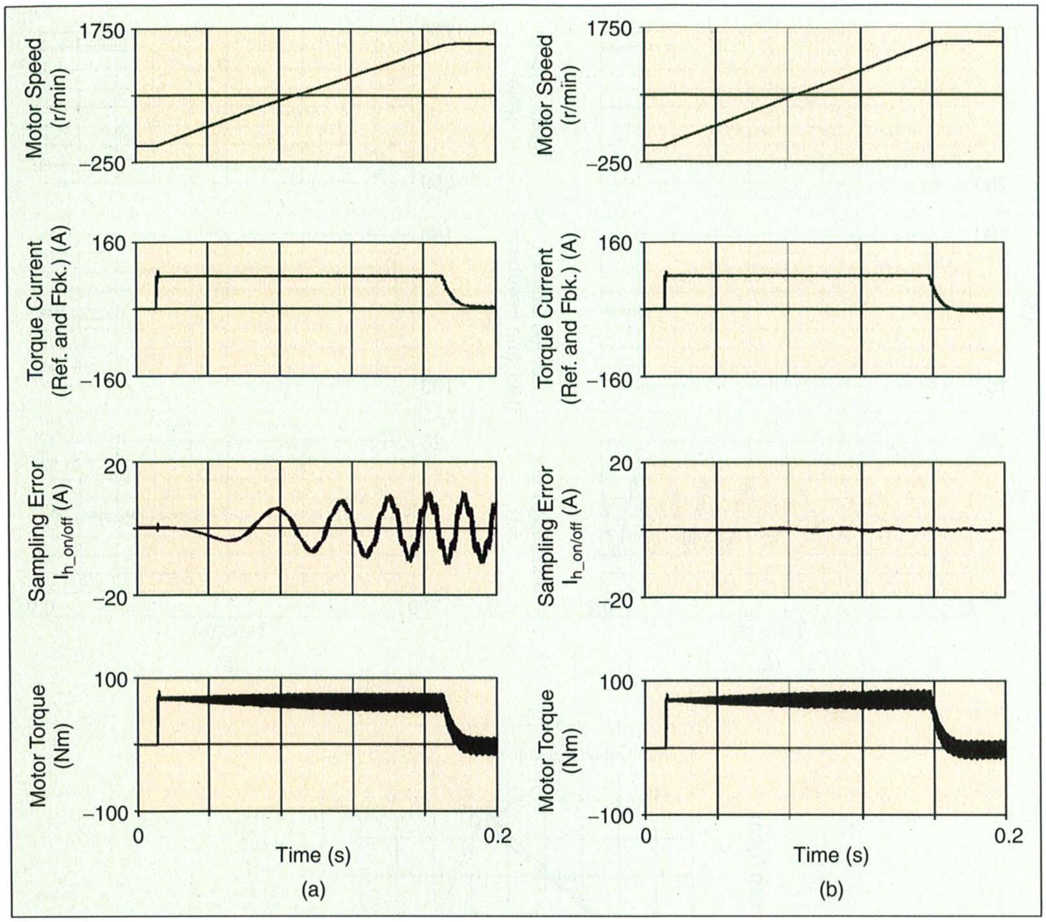

Fig. 11. Torque generation characteristics during acceleration (simulation) (a) without delayed sampling, (b) with delayed sampling. From the top, the motor speed, the reference/feedback of torque current, the current-sampling error, and the reference/real torque.

Conclusions

In this article, the current-sampling error has been investigated in the field-oriented control of an ac machine by a PWM inverter. The expression of the harmonic current due to a low-pass filter has been derived analytically from the reference voltage vector in the SVPWM strategy. The fundamental components of the current-sampling error causes improper torque regulation while the high-frequency ripple of the error introduces lower permissible gain of the current regulators with a resultant impairment in dynamic behavior and limited stability range of the control systems. It has been proposed that the magnitude and phase error from the filter can be avoided with delayed sampling. The effectiveness of the delayed sampling was verified through simulations and experimental results including steady state and dynamic responses.

Appendix

A. Motor

- 11-kW induction motor for spindle application,
- Four-pole, base speed 1500 rpm, maximum speed 6000 rpm,
- Rs=0.04 Ω, Rr=0.0175 Ω, Ls=6.6 mH, Lr=6.6 mH, Lm=6.45 mH.

B. Inverter

- dc link voltage 310 V, IGBT PWM inverter,
- Switching frequency 2.5 kHz, sampling frequency 5 kHz.

C. Current Regulator and PWM Strategy

- Back-EMF compensated PI regulator on synchronously rotating reference frame,
- Fully digital 5-kHz sampling, space vector three-phase symmetry PWM.

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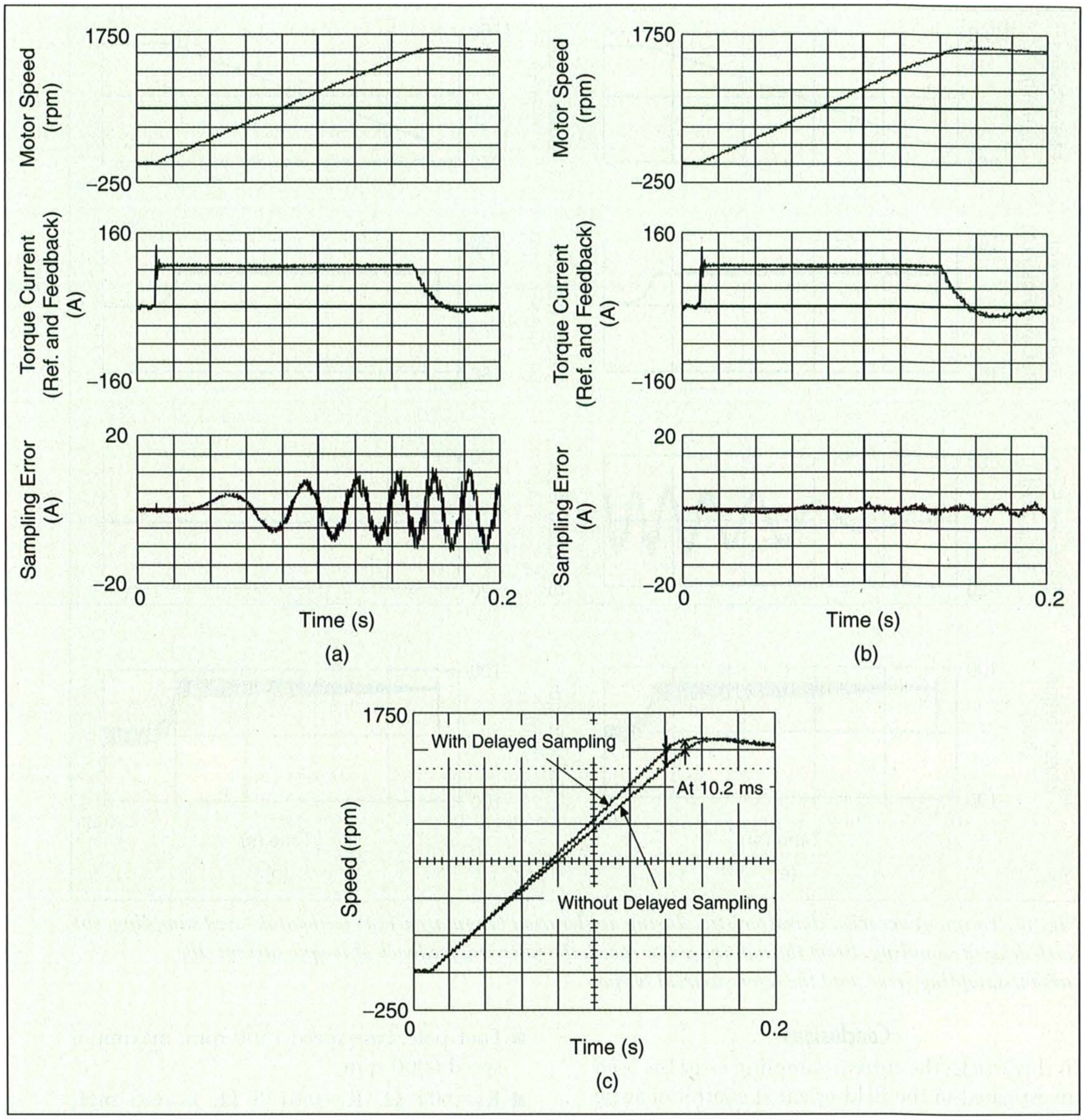

Fig. 12. Torque generation characteristics during acceleration (experiment) (a) without delayed sampling, (b) with delayed sampling. From the top, the motor speed, the reference/feedback of torque current, and the current-sampling error. (c) Comparison of speed responses.

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